

Newton's method for local minimum

$$\frac{\partial f}{\partial \theta}$$

Quadratic approximation: $f(\theta + h) \approx f(\theta) + f'(\theta)h + \underbrace{\frac{1}{2}f''(\theta)h^2}_{g(h)}$

(truncated Taylor series)

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

minimize $g(h)$: condition $\frac{dg}{dh} = f'(\theta) + f''(\theta)h = 0$ gives $h = -\frac{f'(\theta)}{f''(\theta)}$

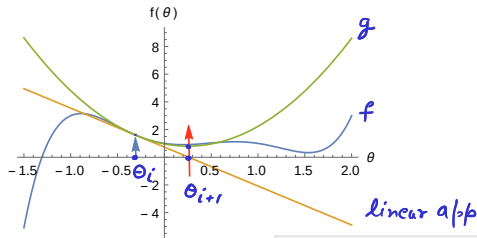
Iteration: $\theta_{i+1} = \theta_i - \frac{f'(\theta_i)}{f''(\theta_i)}$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$\theta_{i+1} = \theta_i - H(\theta_i)^{-1} \nabla f(\theta_i)$$

Hessian matrix: $H = \begin{pmatrix} \frac{\partial^2 f}{\partial \theta_1^2} & \cdots & \frac{\partial^2 f}{\partial \theta_1 \partial \theta_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial \theta_n \partial \theta_1} & \cdots & \frac{\partial^2 f}{\partial \theta_n^2} \end{pmatrix} = \frac{\partial^2 f}{\partial \theta_i \partial \theta_j}$

Gradient vector: $\nabla f = \begin{pmatrix} \frac{\partial f}{\partial \theta_1} \\ \vdots \\ \frac{\partial f}{\partial \theta_n} \end{pmatrix}$



video